THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** second	Second.
2	**(c)** distance / time	Speed formula.
3	**(c)** uniform	Uniform.
4	**(c)** 60 seconds	60 s.
5	**(c)** m/s	Metre per second.
6	**(b)** non-uniform	Non-uniform.
7	**(c)** 60 minutes	60 min.
8	**(a)** 50 km/h	100/2.
9	**(b)** non-uniform motion	Stops and starts.
10	**(b)** 24 hours	24 h.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance per unit time (1); Unit is m/s (1).	1+1
12	Time for one oscillation (1); In seconds (1).	1+1
13	Unequal distance in equal time (1); Changing speed (1).	1+1
14	Second (1); Symbol s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Second (1); Minute (1); Hour (1).	1+1+1
16	Distance = speed x time (1); 40 x 2 (1); 80 km (1).	1+1+1
17	Second hand (1); Fan blades (1); Escalator (1).	1+1+1
	OR: Sundial (1); Water clock (1); Quartz clock (1).	1+1+1
18	Time = distance / speed (1); 150/50 (1); 3 h (1).	1+1+1
	OR: On y-axis (1); With units (1); Against time (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Speed = distance / time (1); 120/20 (1); 6 (1); m/s (1); 6 m/s (1).	1+1+1+1+1
	OR: 18 x 5/18 (1); 5 m/s (1); 10 x 18/5 (1); 36 km/h (1); Use the factor (1).	1+1+1+1+1
20	Sundial (1); Water clock (1); Sand timer (1); Pendulum clock (1); Quartz watch (1).	1+1+1+1+1
	OR: 360/6 (1); 60 km/h (1); 60 x 2 (1); 120 km (1); Uniform speed (1).	1+1+1+1+1

Mistakes to avoid:

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5 m/s	10/2.
2	**(b)** uniform speed	Uniform.
3	**(d)** oscillation	Oscillation.
4	**(c)** 60 km/h	60/1.
5	**(a)** rest	No change in distance.
6	**(a)** time period	Period.
7	**(b)** 60 km/h	240/4.
8	**(d)** x-axis (horizontal)	Horizontal.
9	**(d)** pendulum or quartz	Time keeping.
10	**(b)** 5 m/s	200/40.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	80/2 (1); 40 km/h (1).	1+1
12	Clock (1); Stopwatch (1).	1+1
13	A bob hung from a string (1); Swings to and fro (1).	1+1
14	20/4 (1); 5 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pendulum (1); Clock hands (1); Swing (1).	1+1+1
16	2 x 60 = 120 min (1); 120 x 60 (1); 7200 s (1).	1+1+1
17	For meaning (1); For comparison (1); For clarity (1).	1+1+1
	OR: Bus in traffic (1); Falling leaf (1); Running dog (1).	1+1+1
18	3 x 60 (1); 180 (1); 180 s (1).	1+1+1
	OR: Count 10 oscillations (1); Note the time (1); Divide by 10 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hang a bob (1); Pull and release (1); Count 20 swings (1); Note time (1); Divide by 20 (1).	1+1+1+1+1
	OR: Reset (1); Start on time (1); Stop on time (1); Avoid delay (1); Repeat (1).	1+1+1+1+1
20	Fan: uniform (1); Clock hand: uniform (1); Bus in traffic: non-uniform (1); Falling leaf: non-uniform (1); Kite: non-uniform (1).	1+1+1+1+1
	OR: Pendulum (1); Swing (1); Clock hand (1); Earth's rotation (1); Moon's orbit (1).	1+1+1+1+1

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- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** y-axis (vertical)	Vertical.
2	**(c)** shadow	Shadow.
3	**(a)** 60 km	30 x 2.
4	**(c)** swinging swing	Repeats.
5	**(c)** speedometer	Speed.
6	**(b)** 150 km	50 x 3.
7	**(a)** falling stone	Does not repeat.
8	**(a)** odometer	Distance.
9	**(c)** 50 m	10 x 5.
10	**(b)** stopwatch	Time.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance travelled (1); In a vehicle (1).	1+1
12	One to and fro swing (1); Of a pendulum (1).	1+1
13	Equal distance in equal time (1); Constant speed (1).	1+1
14	Speed (1); In a vehicle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed = distance / time (1); 90/3 (1); 30 km/h (1).	1+1+1
16	Races (1); Experiments (1); Cooking (1).	1+1+1
17	Speed = distance / time (1); Distance = speed x time (1); Time = distance / speed (1).	1+1+1
	OR: 300/60 (1); 5 (1); 5 m/s (1).	1+1+1
18	20/10 (1); 2 (1); 2 s (1).	1+1+1
	OR: m/s (1); km/h (1); cm/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total 120 km (1); Time 2 h (1); Average 120/2 (1); 60 (1); 60 km/h (1).	1+1+1+1+1
	OR: Travel time (1); Safety (1); Racing (1); Fuel use (1); Planning (1).	1+1+1+1+1
20	Axes labelled (1); Points (1,20), (2,40) (1); (3,60), (4,80) (1); Join points (1); Straight line (1).	1+1+1+1+1
	OR: 1 min = 60 s (1); 1 h = 60 min (1); 1 day = 24 h (1); 1 year = 365 days (1); SI unit is second (1).	1+1+1+1+1

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